MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 6**

Inverse functions and exponential functions, and their derivatives (Sections 6.1–6.2, Lectures 11–12); logarithms, their derivatives and logarithmic differentiation (Sections 6.3–6.4, Lecture 13).

Key facts to have at hand (Lectures 11–13)

- **One-to-one:** f never takes the same value twice ($x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$; horizontal line test). Then f has an *inverse*: $f^{-1}(y) = x \iff f(x) = y$. The domain of f^{-1} is the range of f, and the range of f^{-1} is the domain of f. *Undoing:* $f^{-1}(f(x)) = x$ for every x in the domain of f, and $f(f^{-1}(y)) = y$ for every y in the range of f. To find f^{-1}: write $y = f(x)$, solve for x, and keep track of the range. The graph of f^{-1} is the graph of f reflected in $y = x$. And f^{-1} is *not* $1/f$.
- **The derivative of an inverse:** if f is one-to-one and differentiable on an open interval and $f'\big(f^{-1}(a)\big) \neq 0$, then $(f^{-1})'(a) = \dfrac{1}{f'\big(f^{-1}(a)\big)}$ — f' is taken at $f^{-1}(a)$, not at a. If $f'(b) = 0$, then f^{-1} has a vertical tangent at $a = f(b)$ and no derivative there.
- **Exponential functions** ($b > 0$): $b^{x+y} = b^x b^y$, $b^{x-y} = b^x / b^y$, $(b^x)^y = b^{xy}$, $(ab)^x = a^x b^x$. For $b > 1$: increasing, $b^x \to \infty$ as $x \to \infty$, $b^x \to 0$ as $x \to -\infty$; for $0 < b < 1$ the other way round; always $b^x > 0$. e is the base whose slope at 0 is 1: $(e^x)' = e^x$; with the chain rule $\big(e^{g(x)}\big)' = e^{g(x)} g'(x)$. e^x is increasing and continuous, with range $(0, \infty)$; $\lim_{x \to -\infty} e^x = 0$, $\lim_{x \to \infty} e^x = \infty$.
- **Logarithms** ($b > 0$, $b \neq 1$): $\log_b x = y \iff b^y = x$ ($x > 0$). Undoing: $\log_b(b^x) = x$ for every x, $b^{\log_b x} = x$ for $x > 0$. Laws ($x, y > 0$): $\log_b(xy) = \log_b x + \log_b y$, $\log_b \frac{x}{y} = \log_b x - \log_b y$, $\log_b(x^r) = r \log_b x$ — and no law for $\log_b(x + y)$. $\ln = \log_e$; $\log_b x = \dfrac{\ln x}{\ln b}$; $b^x = e^{x \ln b}$. For $b > 1$: $\lim_{x \to 0^+} \log_b x = -\infty$, $\lim_{x \to \infty} \log_b x = \infty$.
- **Derivatives:** $(\ln x)' = \dfrac{1}{x}$ ($x > 0$); $(\ln |x|)' = \dfrac{1}{x}$ ($x \neq 0$); $\big(\ln |g(x)|\big)' = \dfrac{g'(x)}{g(x)}$ where $g(x) \neq 0$; $(b^x)' = b^x \ln b$; $(\log_b x)' = \dfrac{1}{x \ln b}$. *Laws first*: expand a logarithm of a product, quotient or power before differentiating.
- **Is the base constant, or the exponent?** $(b^n)' = 0$; $\big([f(x)]^n\big)' = n[f(x)]^{n-1} f'(x)$; $\big(b^{g(x)}\big)' = b^{g(x)} \ln b \cdot g'(x)$; for $[f(x)]^{g(x)}$ ($f(x) > 0$): *logarithmic differentiation* — take ln of both sides, expand by the laws, differentiate implicitly, solve for y' — or write it as $e^{g(x) \ln f(x)}$. (Logarithmic differentiation also turns products and quotients into sums; where y can be negative, take $\ln |y|$.) And $e = \lim\limits_{x \to 0} (1 + x)^{1/x} = \lim\limits_{n \to \infty} \left(1 + \dfrac{1}{n}\right)^n$.

Part I · Inverse and exponential functions (6.1–6.2) · 50 minutes

Warm-up I · True or false? (justify with one sentence or one counterexample)

1. If f is one-to-one, then $f^{-1}(x) = \dfrac{1}{f(x)}$.
2. The function $f(x) = x^2 + 1$, $x \geq 0$, is one-to-one.
3. $\dfrac{d}{dx} e^{3x} = e^{3x}$.
4. The function $f(x) = x^4$ has an inverse function on $\mathbb{R}$.
5. If f is one-to-one, the domain of f^{-1} is the range of f.
6. If f is one-to-one and differentiable on $\mathbb{R}$, $f(3) = 7$ and $f'(3) = 4$, then $(f^{-1})'(7) = \dfrac{1}{4}$.
7. $(f^{-1})'(a) = \dfrac{1}{f'(a)}$ for every one-to-one differentiable function f.
8. $e^{x+y} = e^x + e^y$ for all real x and y.
9. $\lim\limits_{x \to -\infty} 3^x = 0$.
10. $\dfrac{d}{dx} e^2 = e^2$.

Solution.

1 False. The inverse undoes f; the reciprocal divides 1 by it — different things. Example: $f(x) = 2x$ is one-to-one and $f^{-1}(x) = \dfrac{x}{2}$ (halving undoes doubling), whereas $\dfrac{1}{f(x)} = \dfrac{1}{2x}$. At $x = 2$: $f^{-1}(2) = 1$ but $\dfrac{1}{f(2)} = \dfrac{1}{4}$.

2 True. If $0 \le a < b$, then $a^2 < b^2$ (multiply $a < b$ by $a \ge 0$ and by $b > 0$: $a^2 \le ab < b^2$), so $a^2 + 1 < b^2 + 1$: f is increasing on $[0, \infty)$, and an increasing function never takes the same value twice.

3 False. e^{3x} is a composition: outside e^u, inside $u = 3x$. The chain rule gives $e^{3x} \cdot 3 = 3e^{3x}$. Among the functions e^{kx}, only e^x ($k = 1$) is its own derivative.

4 False. $f(-1) = 1 = f(1)$: two different numbers have the same value, so f is not one-to-one, and "$f^{-1}(1)$" would have to be both -1 and 1. (On $[0, \infty)$ alone, x^4 *is* one-to-one, with inverse $\sqrt[4]{x}$.)

5 True. f^{-1} is defined exactly at the values y that f takes, i.e. on the range of f; and its values are the numbers x of the domain of f. Domain and range swap.

6 True. $f(3) = 7$ means $f^{-1}(7) = 3$, so $(f^{-1})'(7) = \dfrac{1}{f'\big(f^{-1}(7)\big)} = \dfrac{1}{f'(3)} = \dfrac{1}{4}$ (and $f'(3) \neq 0$, so the theorem applies).

7 False. The derivative of f must be taken at $f^{-1}(a)$, not at a. Example: $f(x) = e^x$, whose inverse is ln. Then $(f^{-1})'(a) = (\ln)'(a) = \dfrac{1}{a}$, whereas $\dfrac{1}{f'(a)} = \dfrac{1}{e^a}$; at $a = 1$: $1 \neq \dfrac{1}{e}$. (For a *linear* f the two happen to agree, because f' is the same number at every point — that is why the mistake is easy to make.)

8 False. The law is $e^{x+y} = e^x\, e^y$ — a sum in the exponent becomes a *product*. At $x = y = 0$: $e^0 = 1$, but $e^0 + e^0 = 2$.

9 True. For a base $b > 1$, $b^x \to 0$ as $x \to -\infty$: here $3^x = \dfrac{1}{3^{-x}}$, and $3^{-x} \to \infty$.

10 False. $e^2 \approx 7.39$ is a number; the function $f(x) = e^2$ is constant, and its derivative is 0. "$(e^x)' = e^x$" is about the function e^x, whose exponent varies.

Problem 1 *Inverse functions: one-to-one, the formula, the domain and the range.*

For each function: show that it is one-to-one, find a formula for its inverse, and give the domain and the range of the function and of its inverse.

(a) $f(x) = \dfrac{2x - 1}{x + 3}$

(b) $g(x) = x^2 + 4x, \;\; x \ge -2$ (complete the square)

(c) $h(x) = \sqrt{5 - 2x}$

Solution.

(a) *One-to-one.* **Proof.** Suppose $f(a) = f(b)$, with $a, b \neq -3$. Cross-multiplying, $(2a - 1)(b + 3) = (2b - 1)(a + 3)$, that is $2ab + 6a - b - 3 = 2ab + 6b - a - 3$, so $7a = 7b$ and $a = b$. So f never takes the same value twice. □ *The inverse.* Write $y = \dfrac{2x - 1}{x + 3}$ and solve for x: $y(x + 3) = 2x - 1$, so $xy - 2x = -1 - 3y$, i.e. $x(y - 2) = -(1 + 3y)$, and $x = \dfrac{1 + 3y}{2 - y}$ — possible exactly when $y \neq 2$. Naming the variable x again: $f^{-1}(x) = \dfrac{1 + 3x}{2 - x}$. *Domains and ranges.* f is defined for $x \neq -3$. The computation shows that every $y \neq 2$ is a value of f (at $x = \frac{1+3y}{2-y}$, which is never -3: that would need $1 + 3y = -6 + 3y$), and $y = 2$ is not, because $f(x) = \dfrac{2(x + 3) - 7}{x + 3} = 2 - \dfrac{7}{x + 3} \neq 2$. So: f has domain $\mathbb{R} \setminus \{-3\}$ and range $\mathbb{R} \setminus \{2\}$; f^{-1} has domain $\mathbb{R} \setminus \{2\}$ and range $\mathbb{R} \setminus \{-3\}$ — swapped. *Check:* $f^{-1}(f(0)) = f^{-1}\big(-\frac{1}{3}\big) = \dfrac{1 - 1}{2 + \frac{1}{3}} = 0$.

(b) Complete the square: $g(x) = x^2 + 4x + 4 - 4 = (x+2)^2 - 4$. *One-to-one.* **Proof.** If $-2 \le a < b$, then $0 \le a+2 < b+2$, so $(a+2)^2 < (b+2)^2$ and $g(a) < g(b)$: g is increasing on $[-2, \infty)$, so it never takes the same value twice. □ Its smallest value is $g(-2) = -4$, and $g(x) \to \infty$, so the range is $[-4, \infty)$. *The inverse.* $y = (x+2)^2 - 4 \iff (x+2)^2 = y+4$, and since $x+2 \ge 0$ we take the non-negative root: $x+2 = \sqrt{y+4}$, $x = -2 + \sqrt{y+4}$. So $g^{-1}(x) = -2 + \sqrt{x+4}$, with domain $[-4, \infty)$ (the range of g) and range $[-2, \infty)$ (the domain of g). (The negative root would give the other half of the parabola, which is not in the domain of g.)

(c) *Domain:* $5 - 2x \ge 0$, i.e. $x \le \frac{5}{2}$: $(-\infty, \frac{5}{2}]$. *Range:* a square root is ≥ 0, and every $y \ge 0$ is attained (at $x = \frac{5-y^2}{2}$): $[0, \infty)$. *One-to-one.* **Proof.** If $h(a) = h(b)$, square both sides: $5 - 2a = 5 - 2b$, so $a = b$. □ *The inverse.* $y = \sqrt{5 - 2x}$ with $y \ge 0$; squaring, $y^2 = 5 - 2x$, so $x = \dfrac{5 - y^2}{2}$. Hence $h^{-1}(x) = \dfrac{5 - x^2}{2}$ *for* $x \ge 0$ — domain $[0, \infty)$, range $(-\infty, \frac{5}{2}]$. Without the restriction $x \ge 0$ the formula would describe a whole parabola, which is not one-to-one; the restriction is part of the answer.

Problem 2 *The derivative of an inverse function — without a formula for the inverse.*

(a) Let $f(x) = x^5 + 2x + 3$. Show, without derivatives, that f is increasing (hence one-to-one). Find $f^{-1}(6)$ and $(f^{-1})'(6)$, and also $(f^{-1})'(3)$.

(b) A function f, one-to-one and differentiable on $\mathbb{R}$, has the values below. Find (i) $(f^{-1})'(5)$; (ii) $(f^{-1})'(2)$; (iii) the equation of the tangent line to $y = f^{-1}(x)$ at $x = 9$.

x	0	2	5
$f(x)$	2	5	9
$f'(x)$	4	3	$\frac{1}{2}$

(c) Let $f(x) = (x-1)^3 + 2$. At which number a is f^{-1} *not* differentiable, and why? Check with a formula for f^{-1}.

Solution.

(a) *Increasing.* **Proof.** Let $a < b$. First, $a^5 < b^5$: if $0 \le a < b$, $b^5 - a^5 = (b-a)(b^4 + b^3a + b^2a^2 + ba^3 + a^4) > 0$, because $b - a > 0$ and the second factor is positive ($b > 0$, $a \ge 0$); if $a < b \le 0$, then $0 \le -b < -a$, so $(-b)^5 < (-a)^5$, i.e. $-b^5 < -a^5$; if $a < 0 < b$, then $a^5 < 0 < b^5$. Second, $2a < 2b$. Adding, $a^5 + 2a + 3 < b^5 + 2b + 3$, i.e. $f(a) < f(b)$. So f is increasing and never takes the same value twice. □ $f^{-1}(6)$. We need the x with $x^5 + 2x + 3 = 6$; by inspection $x = 1$ works ($1 + 2 + 3 = 6$), and it is the only one because f is one-to-one. So $f^{-1}(6) = 1$. *The derivative.* $f'(x) = 5x^4 + 2$, never 0. By the theorem, $(f^{-1})'(6) = \dfrac{1}{f'\big(f^{-1}(6)\big)} = \dfrac{1}{f'(1)} = \dfrac{1}{5+2} = \dfrac{1}{7}$. Likewise $f(0) = 3$ gives $f^{-1}(3) = 0$ and $(f^{-1})'(3) = \dfrac{1}{f'(0)} = \dfrac{1}{2}$. We never needed a formula for f^{-1} — solving $x^5 + 2x + 3 = y$ for x is beyond our algebra.

(b) Read the table backwards: $f(0) = 2$, $f(2) = 5$, $f(5) = 9$ mean $f^{-1}(2) = 0$, $f^{-1}(5) = 2$, $f^{-1}(9) = 5$. (i) $(f^{-1})'(5) = \dfrac{1}{f'\big(f^{-1}(5)\big)} = \dfrac{1}{f'(2)} = \dfrac{1}{3}$. (ii) $(f^{-1})'(2) = \dfrac{1}{f'\big(f^{-1}(2)\big)} = \dfrac{1}{f'(0)} = \dfrac{1}{4}$. (The tempting $\dfrac{1}{f'(2)} = \dfrac{1}{3}$ takes f' at the wrong number.) (iii) The point is $\big(9, f^{-1}(9)\big) = (9, 5)$ and the slope is $(f^{-1})'(9) = \dfrac{1}{f'(5)} = \dfrac{1}{1/2} = 2$. Tangent: $y - 5 = 2(x - 9)$, that is $y = 2x - 13$. (Slopes turn upside down: the graph of f has slope $\frac{1}{2}$ at $(5, 9)$, its reflection has slope 2 at $(9, 5)$.)

(c) f is one-to-one, because x^3 is and f is x^3 shifted. $f'(x) = 3(x-1)^2$ is 0 only at $x = 1$, where $f(1) = 2$. The theorem needs $f'\big(f^{-1}(a)\big) \ne 0$, which fails exactly for $a = f(1) = 2$: the graph of f has a horizontal tangent at $(1, 2)$, so its reflection has a *vertical* tangent at $(2, 1)$, and f^{-1} is not differentiable at 2. *Check.* $y = (x-1)^3 + 2 \iff x = 1 + \sqrt[3]{y-2}$, so $f^{-1}(x) = 1 + \sqrt[3]{x-2}$ and, for

$x \neq 2$, $(f^{-1})'(x) = \dfrac{1}{3\left(\sqrt[3]{x-2}\right)^2}$, which grows without bound as $x \to 2$. At 2 itself, from the definition:

$$\lim_{h\to 0} \frac{f^{-1}(2+h) - f^{-1}(2)}{h} = \lim_{h\to 0} \frac{\sqrt[3]{h}}{h} = \lim_{h\to 0} \frac{1}{\left(\sqrt[3]{h}\right)^2} = \infty :$$

no derivative.

Problem 3 *Limits of exponential functions.*

(a) $\lim\limits_{x\to\infty} \dfrac{3e^{2x}+5}{e^{2x}-4}$ and $\lim\limits_{x\to-\infty} \dfrac{3e^{2x}+5}{e^{2x}-4}$

(b) $\lim\limits_{x\to 3^+} 2^{1/(x-3)}$ and $\lim\limits_{x\to 3^-} 2^{1/(x-3)}$

(c) $\lim\limits_{x\to\infty} 5^{-x}\sin x$

(d) $\lim\limits_{x\to\infty} \dfrac{2^x+3^x}{3^x-2^x}$

Solution.

(a) *At* $+\infty$, $e^{2x} \to \infty$ and both top and bottom grow without bound; divide top and bottom by the dominant term e^{2x}:

$$\lim_{x\to\infty} \frac{3e^{2x}+5}{e^{2x}-4} = \lim_{x\to\infty} \frac{3+5e^{-2x}}{1-4e^{-2x}} = \frac{3+0}{1-0} = 3,$$

because $e^{-2x} = \left(e^{-2}\right)^x$ is an exponential with base $0 < e^{-2} < 1$, so it tends to 0 as $x \to \infty$. *At* $-\infty$, $e^{2x} \to 0$ by itself, and there is nothing to divide by:

$$\lim_{x\to-\infty} \frac{3e^{2x}+5}{e^{2x}-4} = \frac{3\cdot 0+5}{0-4} = -\frac{5}{4}.$$

The same function has two different horizontal asymptotes, $y = 3$ and $y = -\frac{5}{4}$.

(b) As $x \to 3^+$, $x-3$ is small and positive, so $\frac{1}{x-3} \to \infty$, and $2^u \to \infty$ as $u \to \infty$. **Proof.** Given $M > 1$, $2^{1/(x-3)} > M$ as soon as $\frac{1}{x-3} > \log_2 M$, i.e. for $3 < x < 3 + \frac{1}{\log_2 M}$. So $\lim\limits_{x\to 3^+} 2^{1/(x-3)} = \infty$. As $x \to 3^-$, $\frac{1}{x-3} \to -\infty$, and $2^u \to 0$ as $u \to -\infty$ (base $2 > 1$): given $\varepsilon \in (0,1)$, $0 < 2^{1/(x-3)} < \varepsilon$ as soon as $\frac{1}{x-3} < \log_2 \varepsilon$, i.e. for $3 - \frac{1}{|\log_2 \varepsilon|} < x < 3$. So $\lim\limits_{x\to 3^-} 2^{1/(x-3)} = 0$. □ The two one-sided limits differ, so the two-sided limit does not exist.

(c) $-1 \le \sin x \le 1$ and $5^{-x} > 0$, so $-5^{-x} \le 5^{-x}\sin x \le 5^{-x}$. Both bounds tend to 0 as $x \to \infty$ (since $5^{-x} = \frac{1}{5^x}$ and $5^x \to \infty$), so by the Squeeze Theorem $\lim\limits_{x\to\infty} 5^{-x}\sin x = 0$.

(d) The dominant term is 3^x. Divide top and bottom by it:

$$\lim_{x\to\infty} \frac{2^x+3^x}{3^x-2^x} = \lim_{x\to\infty} \frac{(2/3)^x+1}{1-(2/3)^x} = \frac{0+1}{1-0} = 1,$$

because $(2/3)^x \to 0$ as $x \to \infty$ (base less than 1).

Problem 4 *Derivatives with e^u.*

Differentiate. For every $e^{\cdots}$ say what the *inside* is before you write anything.

(a) $y = e^{x^3-3x}$

(b) $y = 2x\,e^{-x/2} + e^{-x/2}$

(c) $y = \dfrac{e^{2x}}{x+3}$

(d) $y = \big(e^x + 3\big)^4$

(e) $y = e^{2x}\cos 3x$

(f) $y = e^{e^x}$

Solution. The rule: $\big(e^u\big)' = e^u \cdot u'$ — the exponential, unchanged, times the derivative of the inside.

(a) Inside $u = x^3 - 3x$, $u' = 3x^2 - 3$: $y' = e^{x^3-3x}(3x^2-3) = 3(x^2-1)e^{x^3-3x}$.

(b) A sum of two terms; in both, e^u has inside $u = -\frac{x}{2}$, with $u' = -\frac{1}{2}$. The first term is a product (product rule), the second is a chain: $y' = 2\,e^{-x/2} + 2x \cdot e^{-x/2} \cdot \big(-\frac{1}{2}\big) + e^{-x/2} \cdot \big(-\frac{1}{2}\big) = 2e^{-x/2} - xe^{-x/2} - \frac{1}{2}e^{-x/2}$. Collecting, $y' = \big(\frac{3}{2} - x\big)e^{-x/2} = \dfrac{1}{2}(3-2x)e^{-x/2}$; since $e^{-x/2} > 0$, the sign of y' is the sign of $3 - 2x$.

(c) Quotient rule, with $\big(e^{2x}\big)' = 2e^{2x}$:

$$y' = \frac{(x+3)\cdot 2e^{2x} - e^{2x}\cdot 1}{(x+3)^2} = \frac{e^{2x}\big(2(x+3)-1\big)}{(x+3)^2} = \frac{e^{2x}(2x+5)}{(x+3)^2}.$$

(d) Outside u^4, inside $u = e^x + 3$ with $u' = e^x$: $y' = 4(e^x+3)^3 \cdot e^x$.

(e) Product rule, with $\big(e^{2x}\big)' = 2e^{2x}$ and $(\cos 3x)' = -3\sin 3x$: $y' = 2e^{2x}\cos 3x + e^{2x}(-3\sin 3x) = e^{2x}(2\cos 3x - 3\sin 3x)$.

(f) Outside e^u, inside $u = e^x$, $u' = e^x$: $y' = e^{e^x} \cdot e^x$ $(= e^{e^x+x}$ by the law of exponents). A chain of two exponentials — not e^{e^x} alone.

Problem 5 *Horizontal tangents and tangent lines.*

(a) Find the points of the curve $y = (x^2-3)e^x$ at which the tangent line is horizontal.

(b) Find the equation of the tangent line to $y = (x+1)e^{-2x}$ at $x = 0$.

(c) For which constants k does $y = e^{kx}$ satisfy $y'' - 5y' + 6y = 0$ for every x?

Solution.

(a) Product rule: $y' = 2x\,e^x + (x^2-3)e^x = e^x(x^2+2x-3) = e^x(x+3)(x-1)$. A horizontal tangent means $y' = 0$. Since $e^x > 0$ for every x, we may divide by it: $(x+3)(x-1) = 0$, so $x = -3$ or $x = 1$. The points: $\big(-3, (9-3)e^{-3}\big) = (-3, 6e^{-3}) \approx (-3, 0.30)$ and $\big(1, (1-3)e\big) = (1, -2e) \approx (1, -5.44)$.

(b) The point: $y(0) = (0+1)e^0 = 1$. The slope: product rule, $y' = 1\cdot e^{-2x} + (x+1)\cdot(-2)e^{-2x} = e^{-2x}(1-2x-2) = -(2x+1)e^{-2x}$, so $y'(0) = -1$. Tangent: $y - 1 = -1\cdot(x-0)$, i.e. $y = 1 - x$.

(c) $y = e^{kx}$ gives $y' = ke^{kx}$ and $y'' = k^2e^{kx}$. Then $y'' - 5y' + 6y = e^{kx}(k^2 - 5k + 6)$. This is 0 for every x exactly when $k^2 - 5k + 6 = 0$ (because $e^{kx} > 0$), i.e. $(k-2)(k-3) = 0$: $k = 2$ or $k = 3$. (So e^{2x} and e^{3x} are solutions of this differential equation.)

Problem 6 *Graphs of exponential functions (homework).*

Without a calculator, sketch each graph and state the domain, the range and the horizontal asymptote(s); start from $y = e^x$ or from $y = b^x$ for a suitable b, and say which transformations you used.

(a) $y = 5^{-x}$

(b) $y = 2 - e^{x+1}$

(c) $y = e^{-|x|}$

(d) $y = e^{2-x} - 1$

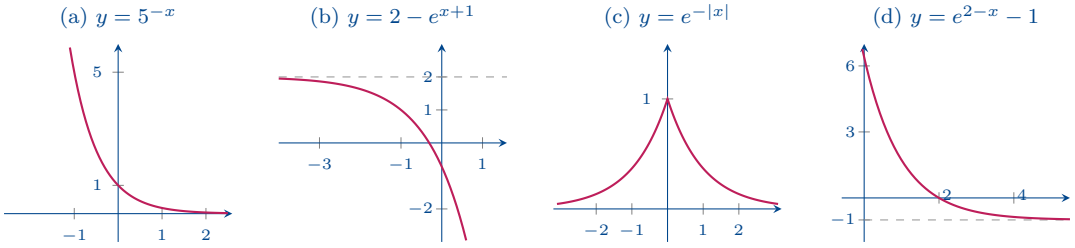
Solution.

(a) $5^{-x} = \left(\frac{1}{5}\right)^x$ is the graph of 5^x reflected in the y-axis. Decreasing; domain $\mathbb{R}$; range $(0, \infty)$; $5^{-x} \to 0$ as $x \to \infty$ (asymptote $y = 0$ on the right) and $\to \infty$ as $x \to -\infty$. Through $(0, 1)$ and $(-1, 5)$.

(b) Start from e^x; shift left by 1 to get e^{x+1} (through $(-1, 1)$); reflect in the x-axis to get $-e^{x+1}$; shift up by 2. Decreasing; domain $\mathbb{R}$; since e^{x+1} takes every value in $(0, \infty)$, $2 - e^{x+1}$ takes every value in $(-\infty, 2)$. As $x \to -\infty$, $e^{x+1} \to 0$: asymptote $y = 2$ on the left. Through $(-1, 1)$; it crosses the x-axis where $e^{x+1} = 2$, at $x = \ln 2 - 1 \approx -0.31$.

(c) For $x \geq 0$, $e^{-|x|} = e^{-x}$; for $x < 0$, $e^{-|x|} = e^x$, the mirror image. So the graph is symmetric in the y-axis (even), with maximum value 1 at $x = 0$, where it has a corner (the slope is 1 from the left and -1 from the right). Domain $\mathbb{R}$; range $(0, 1]$; asymptote $y = 0$ as $x \to \infty$ and as $x \to -\infty$.

(d) $e^{2-x} = e^{-(x-2)}$ is the graph of e^{-x} shifted right by 2 (through $(2, 1)$); then shift down by 1. Decreasing; domain $\mathbb{R}$; range $(-1, \infty)$; as $x \to \infty$, $e^{2-x} \to 0$: asymptote $y = -1$ on the right. Through $(2, 0)$ and $(0, e^2 - 1) \approx (0, 6.39)$.



Part II · Logarithms; logarithmic differentiation (6.3–6.4) · 50 minutes

Warm-up II · True or false? (justify with one sentence or one counterexample)

1. $\log_2 32 = 5$.
2. $\ln(a + b) = \ln a + \ln b$ for all $a, b > 0$.
3. $\log_3 \frac{1}{9} = -2$.
4. $\ln(e^x) = x$ for every real x.
5. $e^{\ln x} = x$ for every real x.
6. $\frac{d}{dx} \ln x = \frac{1}{x}$ for every $x \neq 0$.
7. For every $x \neq 0$, $\frac{d}{dx} \ln(x^2) = \frac{2}{x}$.
8. For every x, $\frac{d}{dx} 5^x = x\, 5^{x-1}$.
9. For $x > 0$, $\frac{d}{dx} x^\pi = \pi\, x^{\pi-1}$.
10. The function $\log_{1/2} x$ is increasing.

Solution.

1 True. $\log_2 32$ is the exponent to which 2 must be raised to give 32: $2^5 = 32$.

2 False. There is no law for the logarithm of a sum. With $a = b = 1$: $\ln(1+1) = \ln 2 \approx 0.69$, while $\ln 1 + \ln 1 = 0$. (The true law is $\ln(ab) = \ln a + \ln b$.)

3 True. $3^{-2} = \frac{1}{9}$, so $\log_3 \frac{1}{9} = -2$.

4 True. ln undoes e^x: $e^x > 0$ for every x, and $\ln(e^x) = x$.

5 False. $\ln x$ is defined only for $x > 0$, so $e^{\ln x}$ makes no sense for $x \le 0$ (for example $x = -1$). For $x > 0$ it is true.

6 False. $\ln x$ itself is defined only for $x > 0$, so it has no derivative at a negative x. The function defined for every $x \neq 0$ with derivative $\frac{1}{x}$ is $\ln|x|$.

7 True. $x^2 = |x|^2$, so $\ln(x^2) = 2\ln|x|$ for every $x \neq 0$ (the law of powers, applied to $|x| > 0$), and $(2\ln|x|)' = \frac{2}{x}$. (Writing $2\ln x$ would lose the negative x — the warm-up of the lecture.)

8 False. In 5^x the base is constant and the exponent varies: $(5^x)' = 5^x \ln 5$. The power rule is for a constant exponent. (The two sides agree only where $x\,5^{x-1} = 5^x \ln 5$, i.e. at $x = 5\ln 5 \approx 8.05$ — not for every x.)

9 True. Here the exponent π is constant and the base varies: the power rule holds for every real exponent, $(x^\pi)' = \pi x^{\pi-1}$ for $x > 0$. (Proof: $x^\pi = e^{\pi \ln x}$ and the chain rule.)

10 False. $\log_{1/2} x = \dfrac{\ln x}{\ln \frac{1}{2}} = -\dfrac{\ln x}{\ln 2} = -\log_2 x$, the mirror image of the increasing function $\log_2 x$ in the x-axis: it is decreasing. (A base between 0 and 1 gives a decreasing logarithm.)

Problem 7 *Derivatives with* ln *— laws first.*

Differentiate. Where the laws of logarithms simplify the function, use them *before* you differentiate.

(a) $y = \ln\big(4 + x^4\big)$

(b) $y = \ln\big[(2x+1)^3(x^2+4)^5\big]$

(c) $y = x^3 \ln(2x), \quad x > 0$

(d) $y = \ln|\sin 2x|$

(e) $y = \log_3\big(5x^2+1\big)$

(f) $y = \dfrac{\ln x}{x^2}$

Solution. The rules: $\big(\ln|u|\big)' = \dfrac{u'}{u}$ — the derivative of the inside over the inside; and $(\log_b u)' = \dfrac{u'}{u \ln b}$.

(a) Inside $u = 4 + x^4 > 0$, $u' = 4x^3$: $y' = \dfrac{4x^3}{4+x^4}$.

(b) *Laws first.* The function is defined where $(2x+1)^3(x^2+4)^5 > 0$, i.e. where $2x+1 > 0$ (the second factor is always positive): $x > -\frac{1}{2}$. There both factors are positive, so the laws apply:

$$y = \ln\big[(2x+1)^3\big] + \ln\big[(x^2+4)^5\big] = 3\ln(2x+1) + 5\ln(x^2+4).$$

Then differentiate term by term: $y' = 3 \cdot \dfrac{2}{2x+1} + 5 \cdot \dfrac{2x}{x^2+4} = \dfrac{6}{2x+1} + \dfrac{10x}{x^2+4}$. (Without the laws: the chain rule around a product of two chains — a page.)

(c) Product rule, with $\big(\ln(2x)\big)' = \dfrac{2}{2x} = \dfrac{1}{x}$ (or: $\ln(2x) = \ln 2 + \ln x$): $y' = 3x^2\ln(2x) + x^3 \cdot \dfrac{1}{x} = x^2\big(3\ln(2x)+1\big)$.

(d) Inside $u = \sin 2x$, $u' = 2\cos 2x$; with the absolute value the formula is the same, wherever $\sin 2x \neq 0$: $y' = \dfrac{2\cos 2x}{\sin 2x} = 2\cot 2x$.

(e) $\log_3 u = \dfrac{\ln u}{\ln 3}$, and $\ln 3$ is a constant: $y' = \dfrac{1}{\ln 3} \cdot \dfrac{10x}{5x^2+1} = \dfrac{10x}{(5x^2+1)\ln 3}$.

(f) Quotient rule: $y' = \dfrac{x^2 \cdot \frac{1}{x} - \ln x \cdot 2x}{x^4} = \dfrac{x - 2x\ln x}{x^4} = \dfrac{1 - 2\ln x}{x^3}$. (So $y' = 0$ at $\ln x = \frac{1}{2}$, i.e. $x = \sqrt{e}$.)

Problem 8 *Logarithmic differentiation.*

(a) $y = \dfrac{\sqrt{x}\,(x+2)^5}{(x^2+4)^2}$, $x>0$: find y', then $y(2)$ and $y'(2)$.

(b) $y = \dfrac{(x-3)^5 e^{2x}}{x^2+1}$: here y can be negative — take $\ln|y|$. Find y' and $y'(0)$.

(c) $y = \dfrac{(x+1)^4\sqrt{2x-1}}{x^2+3}$, $x>\frac{1}{2}$: find y', then $y(1)$ and $y'(1)$.

(d) $y = x^3(x-4)^2$: take $\ln|y|$ and find y'; then find *all* the points of the curve at which the tangent line is horizontal.

Solution. The three steps: (1) take ln of both sides and expand by the laws; (2) differentiate implicitly — the left side gives $\dfrac{y'}{y}$; (3) solve for y' and replace y by the function.

(a) For $x>0$ every factor is positive, so $y>0$ and

$$\ln y = \tfrac{1}{2}\ln x + 5\ln(x+2) - 2\ln(x^2+4).$$

Differentiate: $\dfrac{y'}{y} = \dfrac{1}{2x} + \dfrac{5}{x+2} - \dfrac{2\cdot 2x}{x^2+4} = \dfrac{1}{2x} + \dfrac{5}{x+2} - \dfrac{4x}{x^2+4}$. Hence

$$y' = \frac{\sqrt{x}\,(x+2)^5}{(x^2+4)^2}\left(\frac{1}{2x} + \frac{5}{x+2} - \frac{4x}{x^2+4}\right).$$

At $x=2$: $y(2) = \dfrac{\sqrt{2}\cdot 4^5}{8^2} = \dfrac{1024\sqrt{2}}{64} = 16\sqrt{2}$, and the bracket is $\dfrac{1}{4}+\dfrac{5}{4}-\dfrac{8}{8} = \dfrac{1}{2}$, so $y'(2) = 8\sqrt{2} \approx 11.31$.
(A difference quotient confirms it: $\dfrac{y(2.001)-y(1.999)}{0.002} \approx 11.31$.)

(b) y is negative for $x<3$, so take $\ln|y|$ (for $x\neq 3$):

$$\ln|y| = 5\ln|x-3| + \ln e^{2x} - \ln(x^2+1) = 5\ln|x-3| + 2x - \ln(x^2+1).$$

Differentiate — $\big(\ln|y|\big)' = \dfrac{y'}{y}$, exactly as for $\ln y$: $\dfrac{y'}{y} = \dfrac{5}{x-3} + 2 - \dfrac{2x}{x^2+1}$, so

$$y' = \frac{(x-3)^5 e^{2x}}{x^2+1}\left(\frac{5}{x-3} + 2 - \frac{2x}{x^2+1}\right).$$

At $x=0$: $y(0) = \dfrac{(-3)^5\cdot 1}{1} = -243$ and the bracket is $-\dfrac{5}{3}+2-0 = \dfrac{1}{3}$, so $y'(0) = -81$. (At $x=3$ itself, $y=0$ and the formula cannot be used — but the product rule gives $y'(3)=0$ there, as the factor $(x-3)^5$ shows.)

(c) For $x>\frac{1}{2}$ every factor is positive, so $y>0$ and

$$\ln y = 4\ln(x+1) + \tfrac{1}{2}\ln(2x-1) - \ln(x^2+3).$$

Differentiate; the inside of $\ln(2x-1)$ is $2x-1$, so that term gives $\frac{1}{2}\cdot\frac{2}{2x-1} = \frac{1}{2x-1}$, and $\big(\ln(x^2+3)\big)' = \frac{2x}{x^2+3}$:

$$\frac{y'}{y} = \frac{4}{x+1} + \frac{1}{2x-1} - \frac{2x}{x^2+3}, \qquad y' = \frac{(x+1)^4\sqrt{2x-1}}{x^2+3}\left(\frac{4}{x+1} + \frac{1}{2x-1} - \frac{2x}{x^2+3}\right).$$

At $x=1$: $y(1) = \dfrac{2^4\cdot\sqrt{1}}{4} = 4$, and the bracket is $2+1-\dfrac{2}{4} = \dfrac{5}{2}$, so $y'(1) = 4\cdot\dfrac{5}{2} = 10$.

(d) $y = x^3(x-4)^2$ is negative for $x<0$, positive for $x>0$ ($x\neq 4$), and 0 at $x=0$ and $x=4$; so take $\ln|y|$, for $x\neq 0, 4$:

$$\ln|y| = 3\ln|x| + 2\ln|x-4|, \qquad \frac{y'}{y} = \frac{3}{x} + \frac{2}{x-4} = \frac{5x-12}{x(x-4)}.$$

Hence, for $x \neq 0, 4$, $y' = x^3(x-4)^2 \cdot \dfrac{5x-12}{x(x-4)} = x^2(x-4)(5x-12)$. *At 0 and 4:* y is a polynomial, so y' is a polynomial too, and two polynomials that agree for every $x \neq 0, 4$ agree everywhere; so $y' = x^2(x-4)(5x-12)$ for *every* x. (The product rule gives the same at once: $y' = 3x^2(x-4)^2 + 2x^3(x-4) = x^2(x-4)(5x-12)$.) *Horizontal tangents:* $y' = 0$ gives $x = 0$, $x = 4$ or $x = \frac{12}{5}$. The points: $(0, 0)$; $(4, 0)$; and $y\big(\frac{12}{5}\big) = \big(\frac{12}{5}\big)^3\big(-\frac{8}{5}\big)^2 = \frac{1728}{125} \cdot \frac{64}{25} = \frac{110592}{3125} \approx 35.39$, i.e. $\big(\frac{12}{5}, \frac{110592}{3125}\big)$. *A warning.* The bracket $\frac{3}{x} + \frac{2}{x-4}$ is 0 only at $x = \frac{12}{5}$. The points $x = 0$ and $x = 4$ are where $y = 0$: there $\ln|y|$ does not exist and the formula $y' = y \cdot (\dots)$ says nothing, yet the tangent is horizontal. Logarithmic differentiation finds y' where $y \neq 0$; the zeros of y need a separate look.

Problem 9 *Is the base constant, or the exponent?*

(a) $y = x^{\sqrt{2}} + \big(\sqrt{2}\big)^x$, $x > 0$

(b) $y = (2x)^x$, $x > 0$.

(c) $y = \Big(1 + \dfrac{1}{x}\Big)^x$, $x > 0$

(d) The equation of the tangent line to $y = (x+1)^x$ at $x = 2$.

Solution.

(a) Two terms of different kinds. In $x^{\sqrt{2}}$ the exponent is constant: the power rule, $\sqrt{2}\,x^{\sqrt{2}-1}$. In $(\sqrt{2})^x$ the base is constant: $(b^x)' = b^x \ln b$, so $(\sqrt{2})^x \ln\sqrt{2} = \frac{1}{2}(\sqrt{2})^x \ln 2$. Together: $y' = \sqrt{2}\,x^{\sqrt{2}-1} + \frac{1}{2}\ln 2\,(\sqrt{2})^x$.

(b) Both the base $2x$ and the exponent x vary: neither rule applies as it stands. $\ln y = x\ln(2x)$; differentiate (product rule on the right, with $\big(\ln(2x)\big)' = \frac{1}{x}$): $\dfrac{y'}{y} = 1 \cdot \ln(2x) + x \cdot \dfrac{1}{x} = \ln(2x) + 1$. So $y' = (2x)^x\big(\ln(2x) + 1\big)$.

(c) $\ln y = x\ln\Big(1 + \dfrac{1}{x}\Big) = x\big[\ln(x+1) - \ln x\big]$ (laws first). Product rule: $\dfrac{y'}{y} = \big[\ln(x+1) - \ln x\big] + x\Big(\dfrac{1}{x+1} - \dfrac{1}{x}\Big) = \ln\Big(1 + \dfrac{1}{x}\Big) + \dfrac{x}{x+1} - 1 = \ln\Big(1 + \dfrac{1}{x}\Big) - \dfrac{1}{x+1}$. So

$$y' = \Big(1 + \frac{1}{x}\Big)^x \left(\ln\Big(1 + \frac{1}{x}\Big) - \frac{1}{x+1}\right).$$

(As $x \to \infty$ the function y itself tends to e — put $t = \frac{1}{x}$ in $e = \lim_{t\to 0^+}(1+t)^{1/t}$ — while $y' \to 0$.)

(d) The point: $y(2) = 3^2 = 9$. Logarithmic differentiation: $\ln y = x\ln(x+1)$, $\dfrac{y'}{y} = \ln(x+1) + \dfrac{x}{x+1}$; at $x = 2$: $y'(2) = 9\Big(\ln 3 + \dfrac{2}{3}\Big) = 9\ln 3 + 6 \approx 15.888$. Tangent: $y - 9 = (9\ln 3 + 6)(x-2)$, i.e. $y = (9\ln 3 + 6)x - 18\ln 3 - 3$.

Problem 10 *Equations with logarithms and exponentials.*

Solve. Check every candidate against the domain of the equation.

(a) $e^{3x-1} = 7$

(b) $\ln(x+2) + \ln(x-1) = \ln 10$

(c) $2^{x+1} = 5^x$

(d) $\log_2 x + \log_4 x = 3$

(e) $e^{2x} - 5e^x + 6 = 0$

Solution.

(a) Take ln of both sides (undoing): $3x - 1 = \ln 7$, so $x = \dfrac{1 + \ln 7}{3} \approx 0.982$.

(b) *Domain first:* both logarithms need positive arguments, $x+2 > 0$ and $x-1 > 0$, i.e. $x > 1$. On the domain, by the law of products, $\ln\big[(x+2)(x-1)\big] = \ln 10$, and since ln is one-to-one, $(x+2)(x-1) = 10$,

i.e. $x^2+x-12=0$, $(x+4)(x-3)=0$. The candidates are $x=3$ and $x=-4$; only $x=3$ is in the domain ($\ln(-2)$ does not exist). *Answer:* $x=3$. Check: $\ln 5+\ln 2=\ln 10$.

(c) Take ln of both sides: $(x+1)\ln 2 = x\ln 5$, so $x(\ln 5-\ln 2)=\ln 2$ and $x=\dfrac{\ln 2}{\ln 5-\ln 2}=\dfrac{\ln 2}{\ln\frac{5}{2}}\approx 0.756$.

(d) Domain $x>0$. Change of base: $\log_4 x=\dfrac{\log_2 x}{\log_2 4}=\dfrac{\log_2 x}{2}$. So $\log_2 x+\frac{1}{2}\log_2 x=3$, $\frac{3}{2}\log_2 x=3$, $\log_2 x=2$, $x=2^2=4$. Check: $\log_2 4+\log_4 4=2+1=3$.

(e) A quadratic in $u=e^x$: $u^2-5u+6=0$, $(u-2)(u-3)=0$, so $e^x=2$ or $e^x=3$: $x=\ln 2$ or $x=\ln 3$. (Both are allowed: e^x takes every positive value. If a root u had been negative or zero, it would have been rejected.)

Problem 11 *The number e as a limit; limits with logarithms.*

(a) Show that $\lim\limits_{x\to 0}(1+3x)^{1/x}=e^3$. (Hint: the derivative of $\ln(1+3x)$ at 0.)

(b) Find $\lim\limits_{n\to\infty}\left(\dfrac{n}{n+1}\right)^n$.

(c) $\lim\limits_{x\to\infty}\big[\ln(2x+1)-\ln(x+3)\big]$

(d) $\lim\limits_{x\to\infty}\dfrac{\ln(x^4)}{\ln(x^2)}$

Solution.

(a) Proof. Let $g(x)=\ln(1+3x)$ (for $x>-\frac{1}{3}$). Then $g(0)=\ln 1=0$ and $g'(x)=\dfrac{3}{1+3x}$, so $g'(0)=3$. By the definition of the derivative,

$$3=g'(0)=\lim_{x\to 0}\frac{\ln(1+3x)-\ln 1}{x}=\lim_{x\to 0}\frac{1}{x}\ln(1+3x)=\lim_{x\to 0}\ln\big((1+3x)^{1/x}\big),$$

the last step by the law of powers. Since e^x is continuous, the limit may be moved inside it:

$$e^3=\exp\Big(\lim_{x\to 0}\ln\big((1+3x)^{1/x}\big)\Big)=\lim_{x\to 0}e^{\ln((1+3x)^{1/x})}=\lim_{x\to 0}(1+3x)^{1/x}. \qquad \square$$

(b) Rewrite: $\left(\dfrac{n}{n+1}\right)^n=\dfrac{1}{\left(\frac{n+1}{n}\right)^n}=\dfrac{1}{\left(1+\frac{1}{n}\right)^n}$. Since $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n=e\neq 0$, the limit law for quotients gives

$$\lim_{n\to\infty}\left(\frac{n}{n+1}\right)^n=\frac{1}{\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n}=\frac{1}{e}\approx 0.368.$$

(Numerically: $n=1000$ gives $\left(\frac{1000}{1001}\right)^{1000}\approx 0.3681$, just above $\frac{1}{e}$.)

(c) Laws first, then the continuity of ln at 2:

$$\lim_{x\to\infty}\big[\ln(2x+1)-\ln(x+3)\big]=\lim_{x\to\infty}\ln\frac{2x+1}{x+3}=\ln\Big(\lim_{x\to\infty}\frac{2+1/x}{1+3/x}\Big)=\ln 2.$$

(Each logarithm alone tends to ∞: "$\infty-\infty$" tells nothing until the laws are used.)

(d) For $x>1$: $\dfrac{\ln(x^4)}{\ln(x^2)}=\dfrac{4\ln x}{2\ln x}=2$. The function is constant for $x>1$, so the limit is 2.

Problem 12 *Inverse functions meet logarithms (reserve).*

(a) Let $f(x) = x^3 + \ln x$, $x > 0$. Show that f is one-to-one, and find $(f^{-1})'(1)$.

(b) Let $g(x) = 2 + e^{2x-1}$. Find g^{-1}, with its domain and range, and find $(g^{-1})'(3)$ in two ways: from your formula, and from $\dfrac{1}{g'\big(g^{-1}(3)\big)}$.

(c) Use the formula for the derivative of an inverse function to show that $(\log_b x)' = \dfrac{1}{x \ln b}$, starting from $(b^x)' = b^x \ln b$ ($b > 0$, $b \neq 1$).

Solution.

(a) **Proof** that f is one-to-one. If $0 < a < b$, then $a^3 < b^3$ and $\ln a < \ln b$ (ln is increasing), so $f(a) < f(b)$: f is increasing, hence one-to-one. □ $f(1) = 1 + 0 = 1$, so $f^{-1}(1) = 1$. $f'(x) = 3x^2 + \dfrac{1}{x}$, $f'(1) = 4 \neq 0$, and $(f^{-1})'(1) = \dfrac{1}{f'(1)} = \dfrac{1}{4}$.

(b) Solve $y = 2 + e^{2x-1}$ for x: $e^{2x-1} = y - 2$, possible exactly when $y > 2$; then $2x - 1 = \ln(y-2)$, $x = \dfrac{1 + \ln(y-2)}{2}$. So $g^{-1}(x) = \dfrac{1 + \ln(x-2)}{2}$, with domain $(2, \infty)$ (the range of g: e^{2x-1} takes every positive value) and range $\mathbb{R}$ (the domain of g). *From the formula:* $(g^{-1})'(x) = \dfrac{1}{2} \cdot \dfrac{1}{x-2}$, so $(g^{-1})'(3) = \dfrac{1}{2}$. *From the theorem:* $g\big(\frac{1}{2}\big) = 2 + e^0 = 3$, so $g^{-1}(3) = \frac{1}{2}$; $g'(x) = 2e^{2x-1}$, $g'\big(\frac{1}{2}\big) = 2$; $(g^{-1})'(3) = \dfrac{1}{g'(1/2)} = \dfrac{1}{2}$. The same.

(c) **Proof.** Let $f(x) = b^x$; it is one-to-one and differentiable, $f'(x) = b^x \ln b \neq 0$, and $f^{-1} = \log_b$. For $a > 0$:

$$(\log_b)'(a) = \frac{1}{f'\big(f^{-1}(a)\big)} = \frac{1}{b^{\log_b a} \ln b} = \frac{1}{a \ln b},$$

because $b^{\log_b a} = a$ (undoing). □

Challenge *(for those who finish early, or at home)*

1. Let $f(x) = \log_x 2$ for $x > 0$, $x \neq 1$ (the base varies!). Find $f'(x)$.
2. Differentiate $y = x^{x^x}$, $x > 0$ (the exponent is x^x).
3. Show that the equation $\ln x = 3 - x$ has exactly one solution, and find an interval of length $\frac{1}{2}$ that contains it.

Solution.

1. Change of base: $\log_x 2 = \dfrac{\ln 2}{\ln x} = \ln 2 \cdot (\ln x)^{-1}$. Chain rule (outside u^{-1}, inside $\ln x$): $f'(x) = \ln 2 \cdot (-1)(\ln x)^{-2} \cdot \dfrac{1}{x} = -\dfrac{\ln 2}{x(\ln x)^2}$.

2. Logarithmic differentiation, twice. First, $(x^x)' = x^x(\ln x + 1)$ (from $\ln(x^x) = x \ln x$). Now $\ln y = x^x \ln x$, and by the product rule

$$\frac{y'}{y} = (x^x)' \ln x + x^x \cdot \frac{1}{x} = x^x(\ln x + 1)\ln x + x^{x-1}, \qquad \text{so} \qquad y' = x^{x^x}\Big(x^x(\ln x + 1)\ln x + x^{x-1}\Big).$$

(At $x = 1$: $y'(1) = 1 \cdot (1 \cdot 1 \cdot 0 + 1) = 1$.)

3. **Proof.** Let $h(x) = \ln x + x - 3$ for $x > 0$; the equation is $h(x) = 0$. *At most one solution:* if $0 < a < b$, then $\ln a < \ln b$ and $a < b$, so $h(a) < h(b)$ — h is increasing, hence one-to-one, and takes the value 0 at most once. *At least one:* h is continuous on $[2, 2.5]$, $h(2) = \ln 2 - 1 \approx -0.31 < 0$ and $h(2.5) = \ln 2.5 - 0.5 \approx 0.42 > 0$, so by the Intermediate Value Theorem there is $c \in (2, 2.5)$ with $h(c) = 0$. □ (Numerically $c \approx 2.208$.)